

1BM130 - 10-Minute Biking City

Group 5

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Research Questions

The concept of the 10-minute cycling city is often promoted as a way to improve accessibility and encourage sustainable mobility. At the start of this project, the focus was on understanding spatial differences in accessibility and cycling usage across the Netherlands, as well as the extent to which the benefits of the 10-minute cycling city are distributed equally among different socioeconomic groups.

Where in the Netherlands does the 10-minute cycling city perform well or poorly in terms of access and local cycling usage, and which municipalities are most relevant for policy intervention?

How do different income groups benefit equally from 10-minute cycling access to essential amenities and how does urbanization affect their local cycling behavior.

During the descriptive analysis, however, patterns emerged that suggested accessibility alone could not fully explain cycling behavior. This led to a deeper interest in the factors that drive travel decisions at the individual level. As a result, the study was expanded with an additional research question that focuses on the relationship between accessibility and actual travel behavior.

How do trip characteristics, travel habits, and amenity availability relate to the travel mode choice?

Summary of Findings

The 10-minute cycling city is a meaningful framework, but good access does not automatically lead to more cycling. The municipalities with the highest access scores (Amsterdam, Haarlem, Delft) are not the same as those with the highest local cycling usage. This mismatch pattern is the most policy-relevant result: it shows that access is only one condition for cycling behaviour.

Municipalities with high access but low usage, such as Heerlen, Schiedam, and 's-Gravenhage, need behavioural or infrastructural interventions. Municipalities with low access scores but relatively high cycling usage, such as Nieuwkoop and Scherpenzeel, would benefit most from investments in amenity proximity and more direct cycling routes.

Income turns out to be less decisive than expected: low-income groups often live in denser neighbourhoods and therefore have comparable or even higher access scores. Urbanisation level is a stronger explanatory factor than income alone.

The predictive model (XGBoost) confirms this: trip distance and trip purpose are the strongest predictors of mode choice, followed by travel habits. Amenity availability plays a role but is secondary.

Datasets and Pre-processing

The analysis combines three datasets. **ODiN 2024** is the national travel survey: respondents report all their trips on one assigned day, including mode, purpose, distance, and household income. **CBS 2024** provides neighbourhood-level information: urbanisation class, low-income share, and distances to amenities. The **SmartwayZ buurt-to-buurt** dataset provides actual cycling network distances between neighbourhoods, used to compute the Bike-10 access score.

ODiN was linked to CBS via four-digit postal codes. For the predictive model, all trips from the same respondent were always kept in the same split (GroupShuffleSplit) to prevent data leakage. Only neighbourhoods with ≥ 200 inhabitants and a valid 'bike10_weighted_score' are retained.

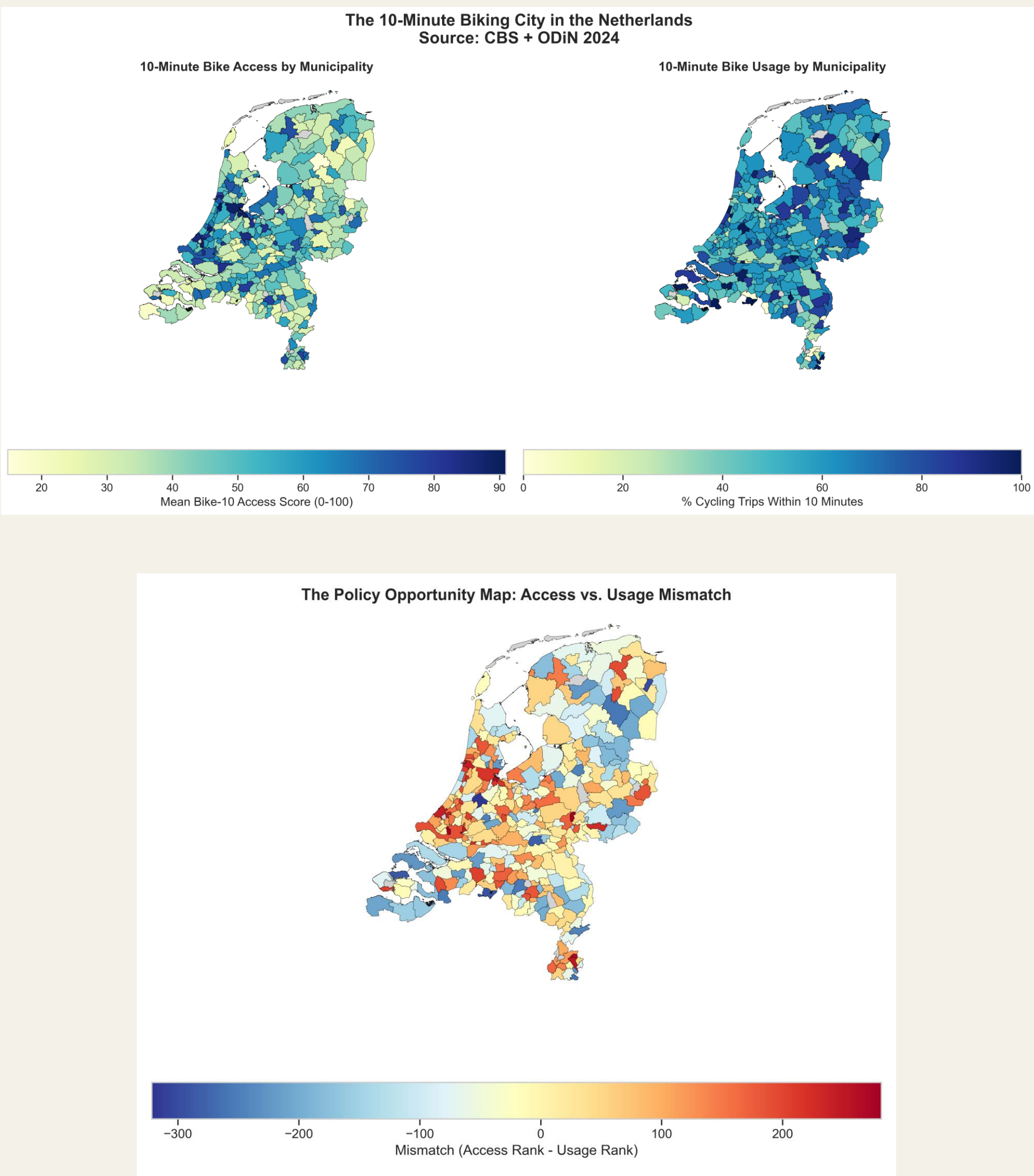
Amenity	Weight
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klasse_huisarts	0.14
klasse_basisschool	0.13
klasse_ziekenhuis	0.10
klasse_apotheek	0.07
klasse_voortgezet_onderwijs	0.07
klasse_bushalte	0.06
klasse_treinstation	0.05
klasse_horeca	0.05
klasse_restaurant	0.04
klasse_sportterrein	0.04
klasse_kinderopvang	0.03
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klasse_kledingwinkel	0.02

Topic 1: Descriptive Analytics

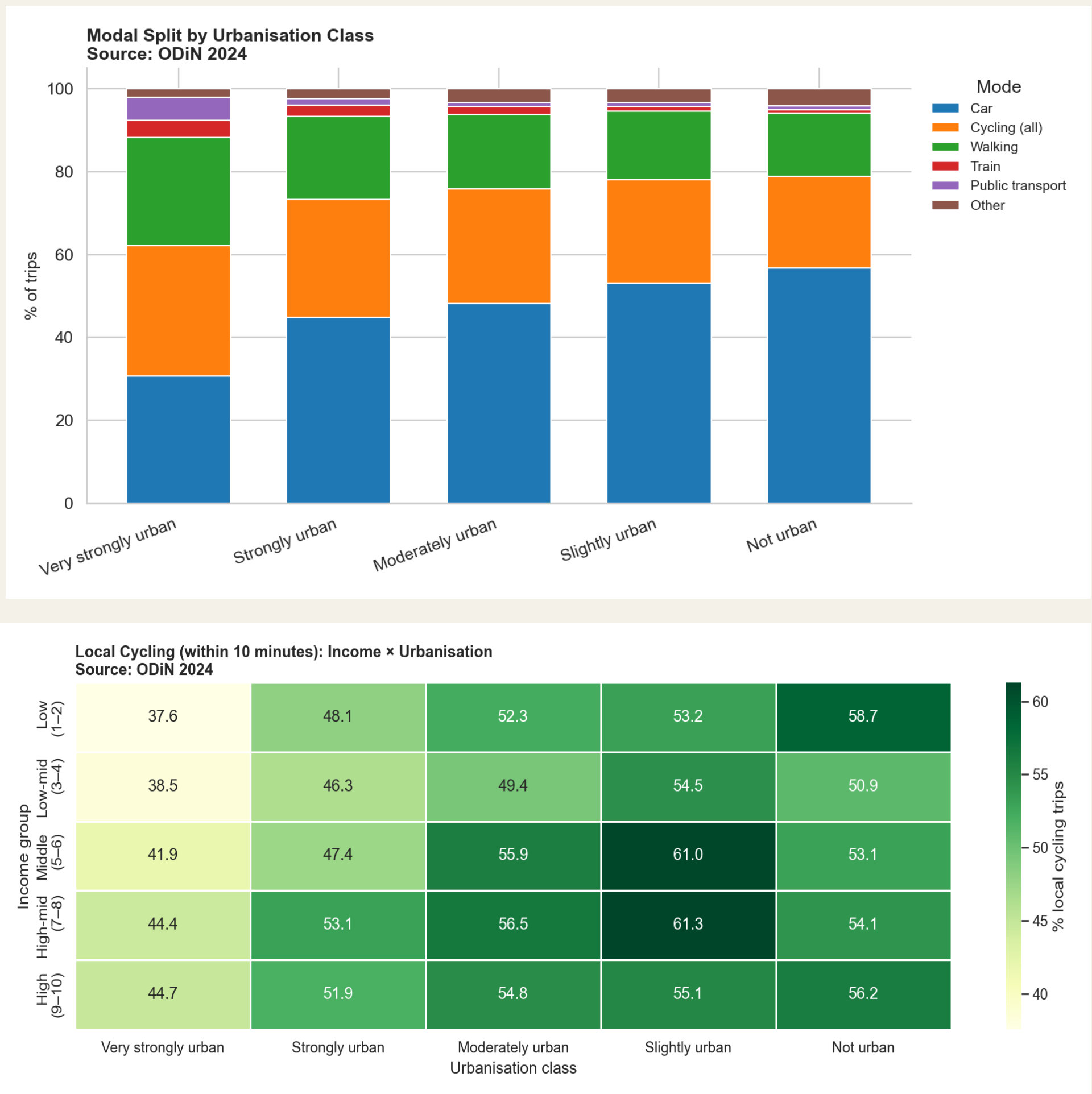
Municipality ranking – access: Highly urbanised municipalities, Amsterdam, Haarlem, Leiden, Delft, have the highest Bike-10 access scores because amenities are clustered close together. Rural municipalities such as Midden-Drenthe and Texel, score lowest: amenities are more dispersed across larger distances.

Municipality ranking - usage: The top-10 on access and the top-10 on usage are two completely different lists. This shows that physical proximity to amenities does not guarantee cycling behaviour. Route attractiveness, traffic safety, and car ownership habits all play a role.

Mismatch gap chart: The mismatch gap (access rank minus usage rank) is the most policy-relevant metric. A positive score means: high access, low usage → opportunity for behavioural intervention. A negative score means: high usage despite low access → opportunity for investments in amenities or more direct cycling infrastructure.



Income × urbanisation heatmap: All income groups show similar shares of local cycling trips once they are already cycling for essential purposes. The difference lies not in income but in urbanisation: in highly urban areas, everyone cycles more locally, regardless of income level.



The study finds that cycling access and usage in Dutch municipalities are shaped more by urbanisation than by income, with highly urban areas enabling more local cycling trips for all income groups. While the 10-minute cycling city concept improves access to essential amenities, actual cycling behavior depends on spatial context, amenity type, and broader mobility factors.

Topic 2: Predictive Analytics

Table 4.1: Recoding of ODiN main transport mode

Original ODiN category	Recoded class
Car driver, car passenger	Car
Train, bus/tram/metro	Public transport
Bicycle	Bike
Walking	Walking

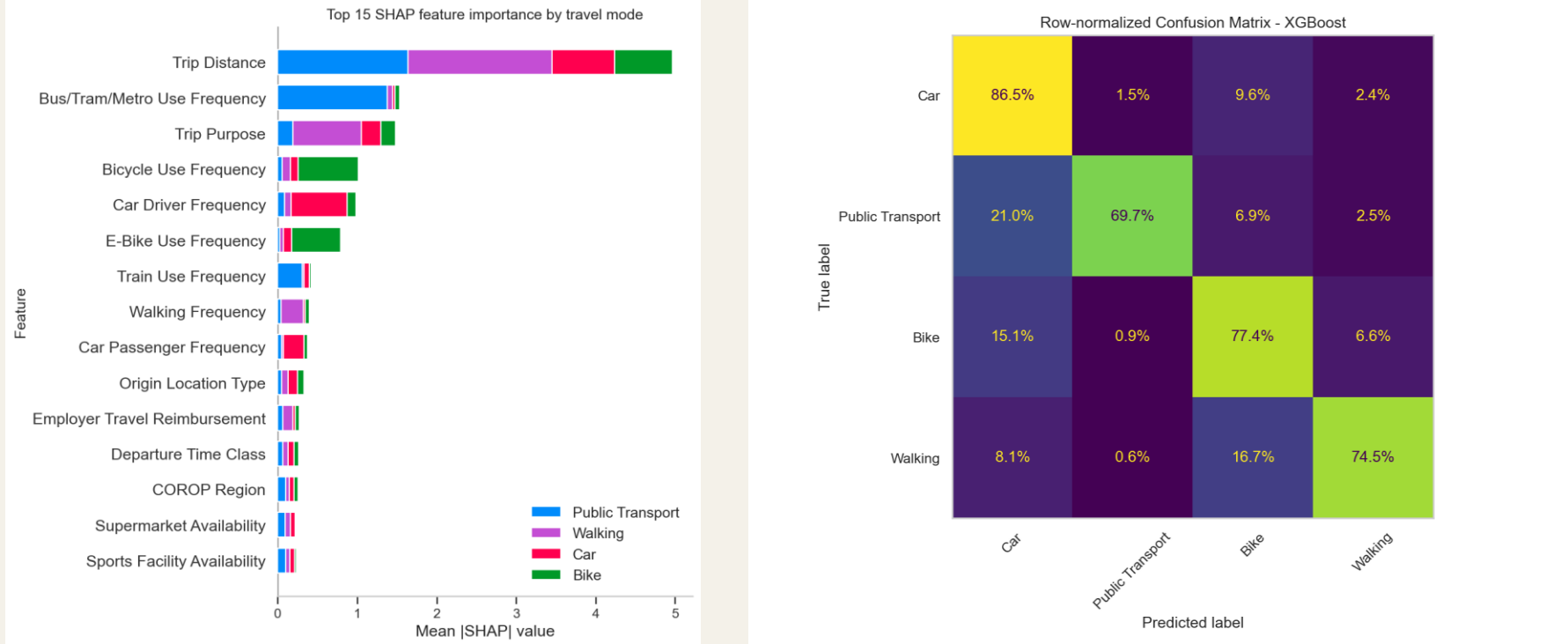
Model overview: An XGBoost model was trained on 172,465 ODiN trips to predict whether someone travels by car, bike, public transport, or on foot. The model achieves 80.5% accuracy and a macro F1-score of 0.78, meaning it performs well across all four transport modes, including the smaller categories.

Table B.1: Hyperparameter search space for Topic 2 models

Model	Hyperparameter	Values searched
Random Forest	Number of trees	200, 400, 600
Random Forest	Maximum depth	None, 10, 20, 30
Random Forest	Minimum samples per leaf	Random integer from 1 to 7
Random Forest	Maximum features	sqrt, log2
Random Forest	Class weighting	balanced, balanced subsample
XGBoost	Number of estimators	250, 400, 600
XGBoost	Maximum depth	3, 4, 5, 6
XGBoost	Learning rate	0.03, 0.05, 0.08, 0.10
XGBoost	Subsample ratio	0.75, 0.90, 1.00
XGBoost	Column subsample ratio	0.75, 0.90, 1.00
XGBoost	Minimum child weight	1, 3, 5, 8
XGBoost	L2 regularisation	1, 3, 5, 8

Confusion matrix: Car trips are predicted most reliably (86.5% correct). Bike and walking trips are sometimes confused (16.7% of walking trips are classified as cycling), which makes sense, as both are used for short distances. Public transport is the hardest to predict (69.7%), reflecting the diversity of train, bus, and tram trips.

SHAP feature importance: Trip distance is by far the strongest predictor: short trips are more likely to be made by foot or bike, long trips by car or public transport. Trip purpose ranks second (commuting differs from shopping), followed by travel habits, people who cycle regularly are more likely to cycle for any given trip. Amenity availability such as supermarket proximity contributes, but is substantially less important than behavioural patterns.



Policy implication: Improving access is necessary but not sufficient. To increase cycling in municipalities that already have good access, behavioural interventions and infrastructure improvements (safety, directness, comfort) are more effective than simply adding more amenities.

Topic 3: The Urban Equity Dashboard

The dashboard is built for municipal policymakers and translates the analytical results into an interactive tool. After selecting a municipality, the user immediately sees: the Bike-10 access score, the low-income share, which neighbourhoods have the weakest access, and how the municipality compares to the national average.

Through the what-if module, a policymaker can simulate what happens when a supermarket, GP, or school is added, or when cycling infrastructure improves by 10–20%. The access score updates immediately to reflect the change.

The Gemini AI assistant at the bottom of the dashboard answers policy-oriented questions in plain language, such as: "What would be the effect of adding a supermarket to the lowest-scoring neighbourhood?" or "What is the best intervention for low-income residents in this municipality?" The assistant is instructed not to make causal claims or invent numbers, it bases all responses solely on the selected municipality's data.

